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# Quartiles and Percentiles

# What is a quartile?



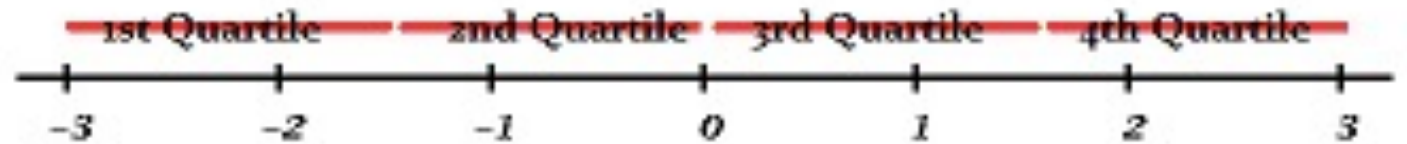
Quartiles in statistics are values that divide your data into quarters. However, quartiles aren't shaped like pizza slices;



Instead they divide your data into four segments according to where the numbers fall on the [number line](#).

# What is a quartile?

- The four quarters that divide a data set into quartiles are:
- The lowest 25% of numbers.
- The next lowest 25% of numbers (up to the [median](#)).
- The second highest 25% of numbers (above the median).
- The highest 25% of numbers.



*A set of numbers (-3,-2,-1,0,1,2,3) divided into four quartiles.*

# What is a quartile?

- As quartiles divide numbers up according to where their position is on the number line, *you have to put the numbers in order before you can figure out where the quartiles are.*

# Steps in getting the quartiles

Quartiles are the values that divide a list of numbers into quarters:

- Put the list of numbers **in order**
- Then cut the list into **four equal parts**
- The Quartiles are at the "cuts"

# Find Quartiles: Example 1

- **Example:** Divide the following data set into quartiles: 2, 5, 6, 7, 10, 22, 13, 14, 16, 65, 45, 12.
- Step 1: Put the numbers in order: 2, 5, 6, 7, 10, 12, 13, 14, 16, 22, 45, 65.
- Step 2: Count how many numbers there are in your set and then divide by 4 to cut the list of numbers into quarters.
- There are 12 numbers in this set, so you would have 3 numbers in each quartile.  
2, 5, 6, | 7, 10, 12 | 13, 14, 16, | 22, 45, 65

# Example # 1: Identifying Q1, Q2, Q3, Q4

\* Q2 is also known as Median

\* Q4 is also known as Maximum

Find the Quartiles for the given data:

2, 3, 3, 4, 4, 6, 7, 7, 7, 8, 9, 11, 12

lower half                      upper half

Q1 - median of  
lower half

Q2  
or  
Median

Q3 - median of  
upper half

$$Q1 = \frac{3+4}{2} = 3.5$$

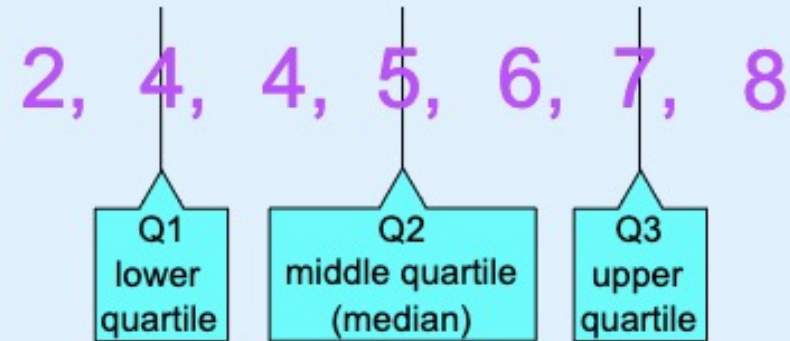
$$Q3 = \frac{8+9}{2} = 8.5$$

# Find Quartiles: Example 2

Example: 5, 7, 4, 4, 6, 2, 8

Put them in order: 2, 4, 4, 5, 6, 7, 8

Cut the list into quarters:



And the result is:

- Quartile 1 (Q1) = **4**
- Quartile 2 (Q2), which is also the Median, = **5**
- Quartile 3 (Q3) = **7**



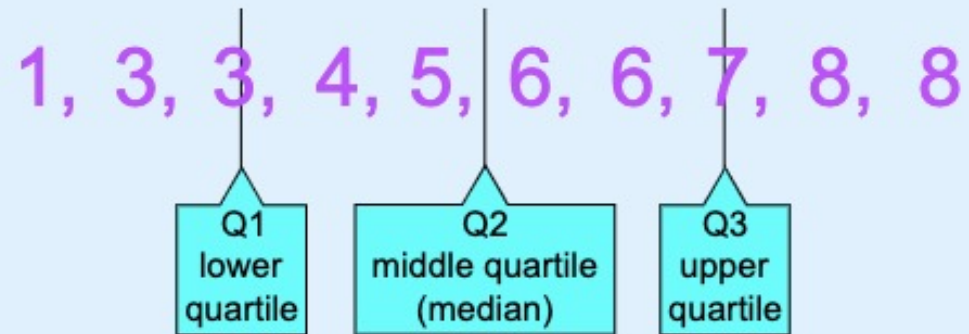
# Find Quartiles: Example 3

- Sometimes a "cut" is between two numbers ... the Quartile is the average of the two numbers.

Example: 1, 3, 3, 4, 5, 6, 6, 7, 8, 8

The numbers are already in order

Cut the list into quarters:



In this case Quartile 2 is half way between 5 and 6:

$$Q2 = (5+6)/2 = \mathbf{5.5}$$

And the result is:

- Quartile 1 (Q1) = **3**
- Quartile 2 (Q2) = **5.5**
- Quartile 3 (Q3) = **7**

# Example:

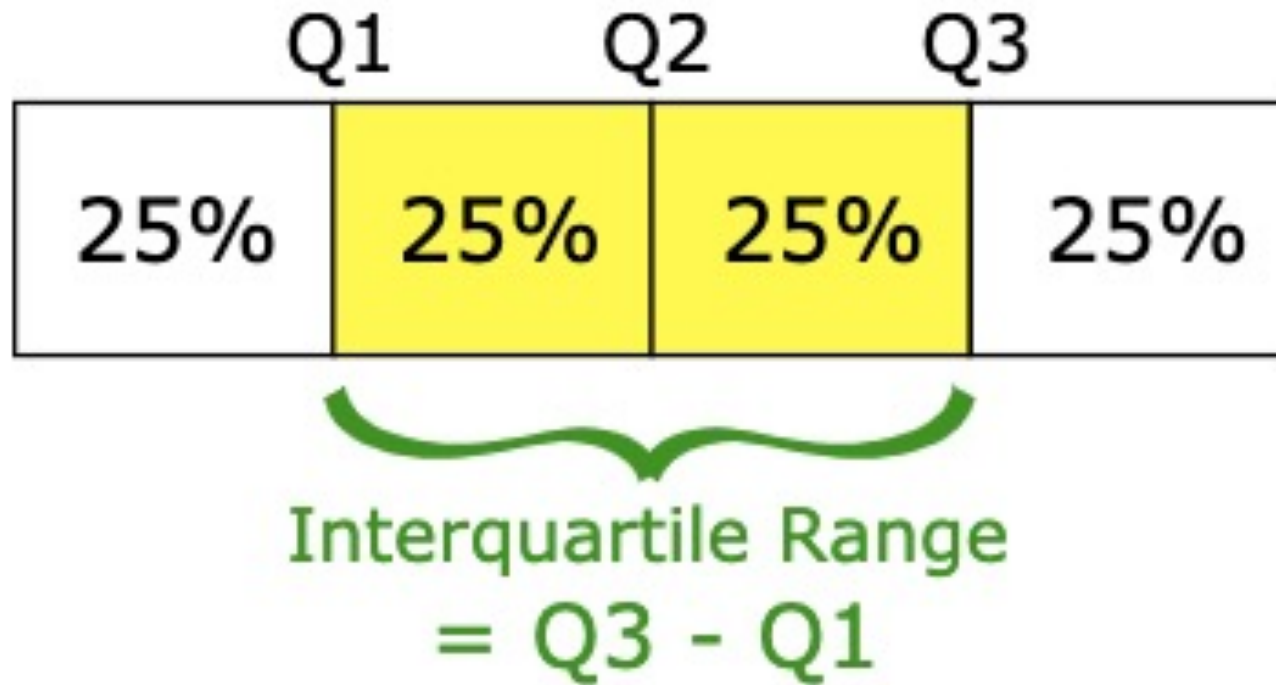
- Compute for Q1, Q2, Q3, Q4 given the following data:  
3, 4, 4, 4, 7, 10, 11, 12, 14, 16, 17, 18

# The Inter Quartile Range (IQR)

- The **interquartile range** is a measure of where the “middle fifty” is in a data set.
- Where a range is a measure of where the beginning and end are in a set, ***an interquartile range is a measure of where the bulk of the values lie.***
- That’s why it’s preferred over many other measures of spread (i.e. the average or median) when reporting things like school performance or SAT scores.
- The interquartile range formula is the first quartile subtracted from the third quartile:
- $IQR = Q_3 - Q_1$ .

# The Inter Quartile Range (IQR)

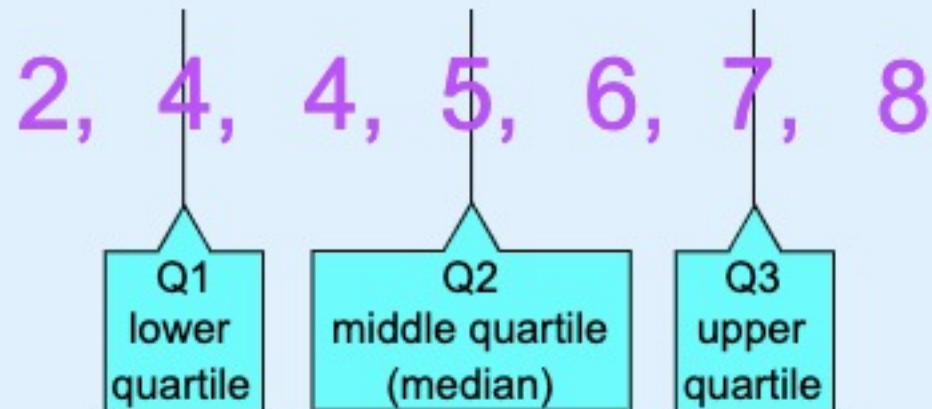
- The "Interquartile Range" is from Q1 to Q3:



# Computing for the Inter Quartile Range (IQR)

- To calculate it just **subtract Quartile 1 from Quartile 3**, like this:

Example:

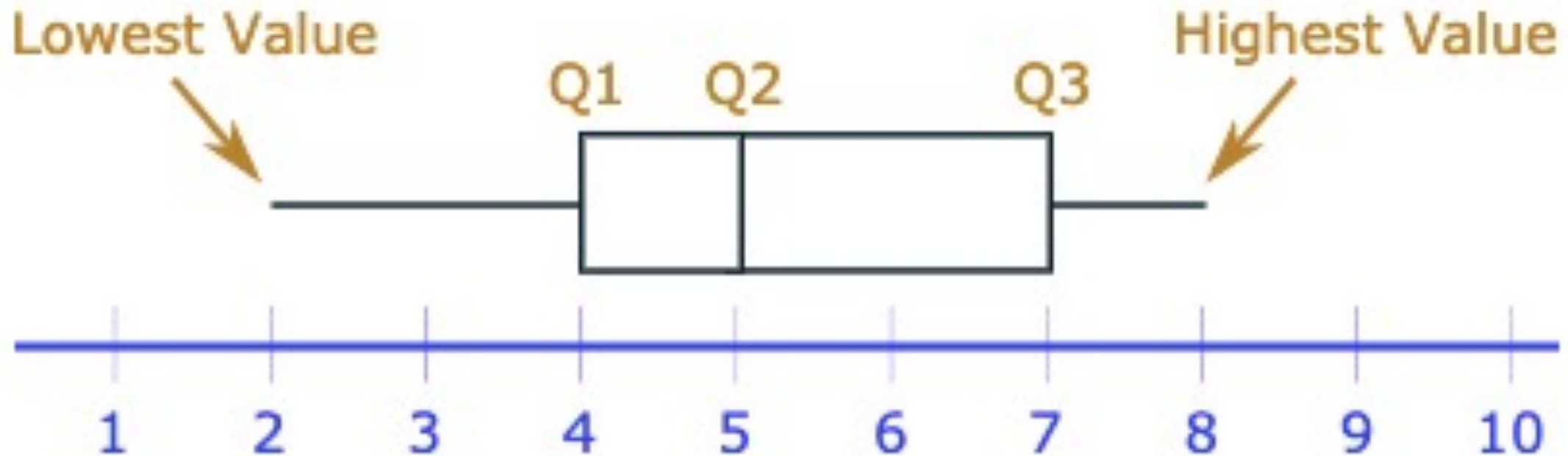


The **Interquartile Range** is:

$$Q3 - Q1 = 7 - 4 = 3$$

## Box and Whisker Plot

We can show all the important values in a "Box and Whisker Plot", like this:



Example: **Box and Whisker Plot and Interquartile Range** for

4, 17, 7, 14, 18, 12, 3, 16, 10, 4, 4, 11

Put them in order:

3, 4, 4, 4, 7, 10, 11, 12, 14, 16, 17, 18

Cut it into quarters:

3, 4, 4 | 4, 7, 10 | 11, 12, 14 | 16, 17, 18

In this case all the quartiles are between numbers:

- Quartile 1 (Q1) =  $(4+4)/2 = \mathbf{4}$
- Quartile 2 (Q2) =  $(10+11)/2 = \mathbf{10.5}$
- Quartile 3 (Q3) =  $(14+16)/2 = \mathbf{15}$

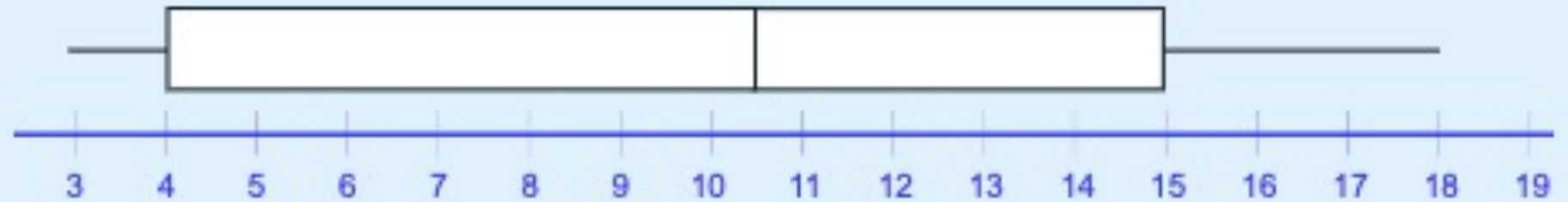
Also:

- The Lowest Value is **3**,
- The Highest Value is **18**



- The Lowest Value is **3**,
- The Highest Value is **18**

So now we have enough data for the **Box and Whisker Plot**:



And the **Interquartile Range** is:

$$Q3 - Q1 = 15 - 4 = \mathbf{11}$$

# Solve

1. What are the quartiles for the following set of numbers?

8, 11, 20, 10, 2, 17, 15, 5, 16, 15, 25, 6

2. What are the quartiles for the following set of numbers?

13, 18, 6, 20, 25, 11, 9, 18, 3, 30, 16, 9, 8, 23, 26, 17

# Percentiles



What is a percentile?

A **percentile** (or a centile) is a measure used in **statistics** indicating the value below which a given percentage of observations in a group of observations falls.

Percentile: the value below which a percentage of data falls.

Example: You are the fourth tallest person in a group of 20

80% of people are shorter than you:



That means you are at the **80th percentile**.

If your height is 1.85m then "1.85m" is the 80th percentile height in that group.

Example

# In Order



Have the data **in order**, so you know which values are above and below.



To calculate percentiles of height: have the data in height order (sorted by height).



To calculate percentiles of age: have the data in age order.



And so on.

# Applications of Percentile

- Percentile scores have a variety of uses. Anytime that a set of data needs to be broken into digestible chunks, percentiles are helpful.
- They are often used to interpret test scores—such as NAT scores—so that test-takers can compare their performance to that of other students.
- *For example, a student might earn a score of 90 percent on an exam. That sounds pretty impressive; however, it becomes less so when a score of 90 percent corresponds to the 20th percentile, meaning only 20 percent of the class earned a score of 90 percent or lower.*

# Formula for Computing a percentile

- Just as there are three quartiles separating a data set into four parts, there are also 99 percentiles, denoted  $P_1, P_2, P_3, \dots, P_{99}$ , which partition the data into 100 groups with about 1% of the values in each group.
- The process of finding the percentile that corresponds to *a particular value  $x$*  is fairly simple, as indicated in the following expression:

$$\text{percentile of value } x = \frac{\text{number of values less than } x}{\text{total number of values}} \cdot 100$$

(round the result to the nearest whole number)



# Example: Ages of Best Actresses

- Table 3-4 lists the 76 sorted ages of Oscar-winning Best Actresses.  
*Find the percentile corresponding to the age of 30 years.*

| Table 3-4      Sorted Ages of 76 Best Actresses |    |    |    |    |    |    |    |    |    |
|-------------------------------------------------|----|----|----|----|----|----|----|----|----|
| 21                                              | 22 | 24 | 24 | 25 | 25 | 25 | 25 | 26 | 26 |
| 26                                              | 26 | 27 | 27 | 27 | 27 | 28 | 28 | 28 | 28 |
| 29                                              | 29 | 29 | 29 | 29 | 29 | 30 | 30 | 31 | 31 |
| 31                                              | 32 | 32 | 33 | 33 | 33 | 33 | 33 | 34 | 34 |
| 34                                              | 35 | 35 | 35 | 35 | 35 | 35 | 35 | 36 | 37 |
| 37                                              | 38 | 38 | 38 | 38 | 39 | 39 | 40 | 41 | 41 |
| 41                                              | 41 | 41 | 42 | 42 | 43 | 45 | 46 | 49 | 50 |
| 54                                              | 60 | 61 | 63 | 74 | 80 |    |    |    |    |

# Example: Ages of Best Actresses

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**SOLUTION** From Table 3-4 we see that there are 26 ages less than 30, so

$$\text{percentile of 30} = \frac{26}{76} \cdot 100 = 34$$

**INTERPRETATION** The age of 30 years is the 34th percentile.

# Reverse Method

- There are several different methods for the reverse procedure of converting a given percentile to the corresponding value in the data set.

## Notation

- $n$  total number of values in the data set
- $k$  percentile being used (Example: For the 25th percentile,  $k = 25$ .)
- $L$  locator that gives the *position* of a value (Example: For the 12th value in the sorted list,  $L = 12$ .)
- $P_k$   $k$ th percentile (Example:  $P_{25}$  is the 25th percentile.)

# Example: Ages of Best Actresses

- Refer to the sorted ages of the Best Actresses in Table 3-4 and use Figure 3-6 to *find the value of the 20th percentile,  $P_{20}$ .*

**SOLUTION** Referring to Figure 3-6, we see that the sample data are already sorted, so we can proceed to find the value of the locator  $L$ . In this computation we use  $k = 20$  because we are trying to find the value of the 20th percentile. We use  $n = 76$  because there are 76 data values.

$$L = \frac{k}{100} \cdot n = \frac{20}{100} \cdot 76 = 15.2$$

# Example: Ages of Best Actresses

- Next, we are asked if  $L$  is a whole number and we answer no, so we proceed to the next lower box where we change  $L$  by *rounding it up* from *15.2 to 16*. (Round up is different from round off).
- Finally, the bottom box shows that the value of  $P_{20}$  is the 16th value, counting from the lowest.
- In Table 3-4, the 16th value is 27. That is  $P_{20} = 27$ , years of age.

# When data is grouped:

When the data is grouped:

- Add up all percentages **below** the score, plus **half** the percentage **at** the score.

## Example: You Score a B!

In the test 12% got D, 50% got C, 30% got B and 8% got A

You got a B, so add up

- all the 12% that got D,
- all the 50% that got C,
- half of the 30% that got B,



for a total percentile of  $12\% + 50\% + 15\% = 77\%$

In other words you did "as well or better than 77% of the class"

(Why take half of B? Because you shouldn't imagine you got the "Best B", or the "Worst B", just an average B.)



# Estimating Percentiles

- We can estimate percentiles from a [line graph](#).

## Example: Shopping

A total of 10,000 people visited the shopping mall over 12 hours:

| Time (hours) | People |
|--------------|--------|
| 0            | 0      |
| 2            | 350    |
| 4            | 1100   |
| 6            | 2400   |
| 8            | 6500   |
| 10           | 8850   |
| 12           | 10,000 |

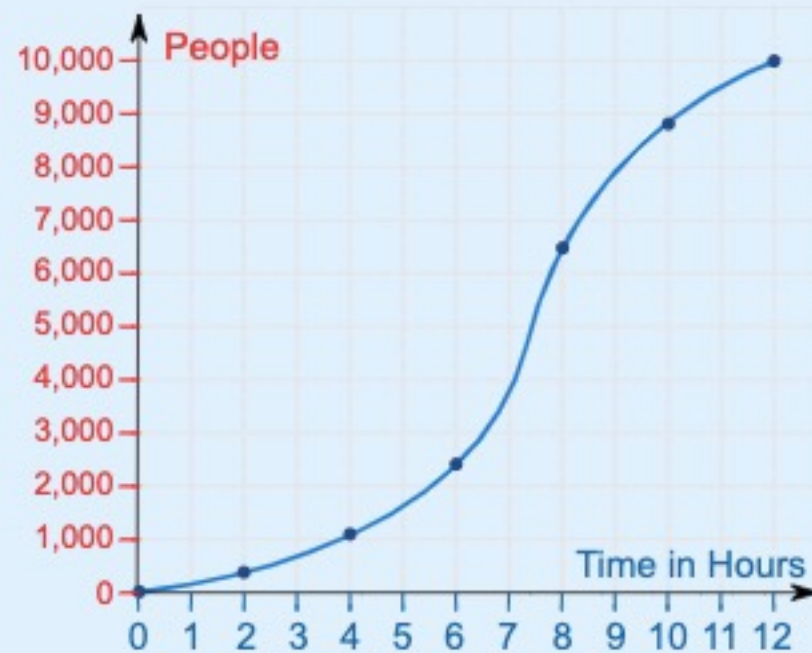


a) Estimate the 30th percentile (when 30% of the visitors had arrived).

b) Estimate what percentile of visitors had arrived after 11 hours.

# Estimating Percentiles

First draw a line graph of the data: plot the points and join them with a smooth curve:

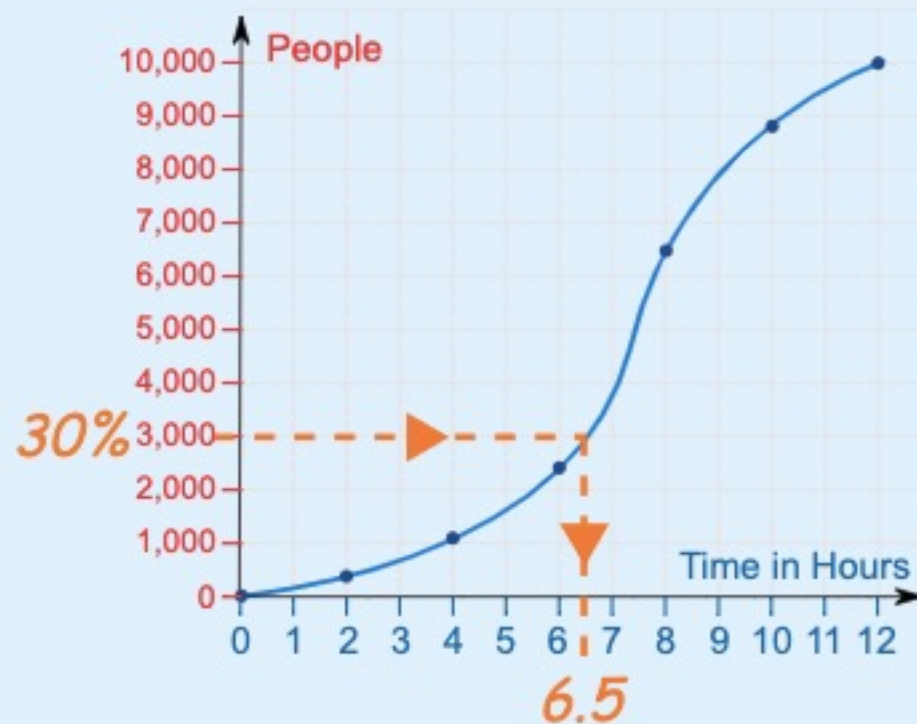


a) The 30th percentile occurs when the visits reach 3,000.



# Estimating Percentiles

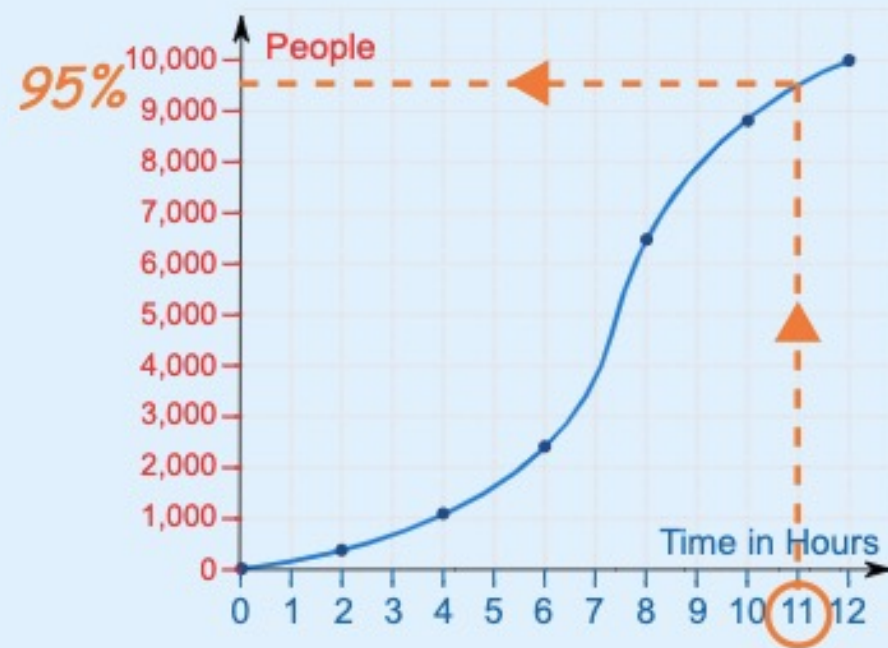
Draw a line horizontally across from 3,000 until you hit the curve, then draw a line vertically downwards to read off the time on the horizontal axis:



So the **30th percentile** occurs after about **6.5 hours**.

# Estimating Percentiles

b) To estimate the percentile of visits after 11 hours: draw a line vertically up from 11 until you hit the curve, then draw a line horizontally across to read off the population on the vertical axis:



So the visits at **11 hours** were about **9,500**, which is the **95th percentile**.